

The Generation of Words with Special Properties

J.S. Lin *

L.C. Juan[†]

R.C.T. Lee[†]

Abstract

We propose some new word generation functions and study his properties. These functions include Fibonacci words, alternating Fibonacci words, the block palindrome words and the suffix-prefix Fibonacci words.

1 Introduction

The word generation functions are interested by many researchers. Especially, Fibonacci word has been studied for many years. The combinatorial properties of some word generation functions are of great interest in some aspects of mathematics and computer science, such as number theory[2], fractal geometry[9], formal language[13], computational complexity[3, 11], string processing[8], data compressing[12] and pattern matching[10].

In this paper, we present some new Fibonacci word properties and some new word generation functions and his properties. These generation function include Fibonacci words, alternating Fibonacci words, the block palindrome words and suffix-prefix Fibonacci words. The palindrome or block palindrome structure of DNA or RNA has

been studied by many authors[1, 15, 15].

The paper is organized as follows: Fibonacci word generation functions and its properties is stated in Section 2. In Section 3, we depict the alternating Fibonacci words generation functions. The block palindrome word is defined in Section 4. In Section 5, we give a new word generation function called suffix-prefix Fibonacci word. Finally, we give a conclusion in Section 6.

2 A Fibonacci Word Generation Functions

An n -Fibonacci word f_n is a word which satisfies the following condition: $f_n = f_{n-1} + f_{n-2}$, or $f_n = f_{n-2} + f_{n-1}$ where $+$ is concatenation operation.

For instance, the following words are Fibonacci words: $f_1 = b$, $f_2 = ab$, $f_3 = bab$, $f_4 = abbab$. Besides, $f_1 = b$, $f_2 = ba$, $f_3 = bab$, $f_4 = babba$ are also Fibonacci words.

Fibonacci words properties is studies by many researchers. Palindromes in the Fibonacci word is discussed in [4, 14, 6]. The exact number of squares in Fibonacci words[7] Locating factors of the infinite Fibonacci word[5]

In his paper, we propose a new word generating function $\phi(f_n)$ as follows:

$$\begin{aligned} \phi(x_1 x_2 \cdots x_m) &= \phi(x_1) \phi(x_2) \cdots \phi(x_m), x_i \in \{a, b\} \\ \phi(a) &= b, \quad \phi(b) = ab, \quad f_0 = a, \quad f_n = \phi(f_{n-1}). \end{aligned}$$

*Department of Information Management, Southern Taiwan University of Technology, js-lin2003@mail.stut.edu.tw

[†]Department of Computer Science and Information Engineering, National Chi-Nan University, rctlee@ncnu.edu.tw

The following words will be generated by the function $\phi(f_n)$:

$$\begin{aligned} f_0 &= a \\ f_1 &= \phi(f_0) = \phi(a) = b \\ f_2 &= \phi(f_1) = \phi(b) = ab \\ f_3 &= \phi(f_2) = \phi(ab) = bab \\ f_4 &= \phi(f_3) = \phi(bab) = abbab \\ f_5 &= \phi(f_4) = \phi(abbab) = bababbab \\ f_6 &= \phi(f_5) = \phi(bababbab) = abbabbababbab \end{aligned}$$

We first show that $\phi(f_n)$ is an n-Fibonacci word generating function..

Property 2.1. $f_n = f_{n-2} + f_{n-1}$, for $n \geq 2$.

Proof. Note that, $f_0 = a$, $f_1 = b$ and $f_2 = ab = f_0 + f_1$. Thus $n = 2$ is valid.

Assume that $f_k = f_{k-2} + f_{k-1}$ for $k = 2, 3, \dots, n$.

Next, consider

$$\begin{aligned} f_{k+1} &= \phi(f_k) \\ &= \phi(f_{k-2} + f_{k-1}) \\ &= \phi(f_{k-2}) + \phi(f_{k-1}) \\ &= f_{k-1} + f_k \end{aligned}$$

Therefore, f_n is a Fibonacci word. $\square$

Before we describe the properties of Fibonacci words generated by ϕ , we define some notations. The k^{th} character of n^{th} word f_n , which is generated by ϕ , is denoted as $c_{n,k}$.

For example: $c_{4,2}$ is b, $c_{5,2}$ is a.

Property 2.2. $c_{n,1} + c_{n,2} = c_{n+1,2} + c_{n+1,1}$ for $n \geq 2$

Property 2.3. The suffix of words generated by ϕ are ab for all n .

The above two properties are obvious, we omit the proof.

Let f_n^s be the suffix of the n^{th} word whose length is $|f_n| - 2$.

Example: $f_5 = bababbab$, $f_5^s = babbab$.

Let $(f_n)^R$ be the reverse word of f_n .

Example: $(f_5)^R = babbabab$, $(f_5^s)^R = babbab$.

Property 2.4. $f_n^s = (f_n^s)^R$ for $n \geq 3$

Proof. (By induction) Note that, $f_3 = bab$ and $f_3^s = b = (f_3^s)^R$, $n=3$ is true.

Assume that $f_k^s = (f_k^s)^R$ is true for $3 \leq k < n$.

We want to show that $n = k$ is also true.

By definition, we have

$$f_k = c_{k,1} + c_{k,2} + f_k^s$$

and

$$f_k = f_{k-2} + f_{k-1}$$

Thus,

$$c_{k,1} + c_{k,2} + f_k^s = c_{k-2,1} + c_{k-2,2} + f_{k-2}^s + f_{k-1}$$

By Property 2.2, we have $c_{k,1} + c_{k,2} = c_{k-2,1} + c_{k-2,2}$.

Therefore,

$$f_k^s = f_{k-2}^s + f_{k-1}$$

and

$$f_k^s = f_{k-2}^s + c_{k-1,1} + c_{k-1,2} + f_{k-1}^s$$

Thus,

$$\begin{aligned} & (f_{k-2}^s + c_{k-1,1} + c_{k-1,2} + f_{k-1}^s)^R \\ &= (f_{k-1}^s)^R + c_{k-1,2} + c_{k-1,1} + (f_{k-2}^s)^R \\ &= (f_{k-1}^s)^R + c_{k-2,1} + c_{k-2,2} + (f_{k-2}^s)^R \end{aligned}$$

(By Property 2.2)

$$\begin{aligned}
 &= f_{k-1}^s + c_{k-2,1} + c_{k-2,2} + f_{k-2}^s \\
 &= f_{k-1}^s + f_{k-2} \\
 &= (f_{k-3}^s + f_{k-2}) + f_{k-2} \\
 &= (f_{k-3}^s + f_{k-4} + f_{k-3}) + f_{k-2} \\
 &= f_{k-3}^s + f_{k-4} + (f_{k-3} + f_{k-2}) \\
 &= f_{k-3}^s + f_{k-4} + f_{k-1}
 \end{aligned}$$

Next, we want to show that $f_{k-2}^s = f_{k-3}^s + f_{k-4}$

By assumption, $f_{k-2}^s = (f_{k-2}^s)^R$, for $3 \leq k < n$

$$\begin{aligned}
 f_{k-2}^s &= (f_{k-2}^s)^R \\
 &= (f_{k-4}^s + f_{k-3})^R \\
 &= (f_{k-4}^s + c_{k-3,1} + c_{k-3,2} + f_{k-3}^s)^R \\
 &= (f_{k-3}^s)^R + c_{k-3,2} + c_{k-3,1} + (f_{k-4}^s)^R \\
 &= (f_{k-3}^s)^R + c_{k-4,1} + c_{k-4,2} + (f_{k-4}^s)^R \\
 &= f_{k-3}^s + (c_{k-4,1} + c_{k-4,2} + f_{k-4}^s) \\
 &= f_{k-3}^s + f_{k-4}
 \end{aligned}$$

Thus,

$$\begin{aligned}
 (f_k^s)^R &= (f_{k-2}^s + c_{k-1,1} + c_{k-1,2} + f_{k-1}^s)^R \\
 &= f_{k-3}^s + f_{k-4} + f_{k-1} \\
 &= f_{k-2}^s + f_{k-1} \\
 &= f_k^s
 \end{aligned}$$

Thus, $f_n^s = (f_n^s)^R$ for $n \geq 3$. $\square$

In the Table 1, we list several other Fibonacci words generating functions. It can be seen that the properties of words generated by ϕ_1 , ϕ_2 and ϕ_3 are quite similar to those of the words generated by function ϕ .

Table 1: The Other Fibonacci Word Generation Functions

ϕ_1	$f_0 = a$	$\phi_1(a) = b$	$\phi_1(b) = ba$	$f_n = \phi_1(f_{n-1})$
ϕ_2	$f_0 = a$	$\phi_2(a) = ab$	$\phi_2(b) = a$	$f_n = \phi_2(f_{n-1})$
ϕ_3	$f_0 = a$	$\phi_3(a) = ba$	$\phi_3(b) = a$	$f_n = \phi_3(f_{n-1})$

3 Alternating Fibonacci Words

In this section, we propose a new class of Fibonacci words, called alternating Fibonacci words. An alternating Fibonacci word is a word which satisfies the condition: $f_n = f_{n-2} + f_{n-1}$ if n is odd and $f_n = f_{n-1} + f_{n-2}$ if n is even, or $f_n = f_{n-1} + f_{n-2}$ if n is odd and $f_n = f_{n-2} + f_{n-1}$ if n is even.

We propose an alternating Fibonacci word function $\varphi(f_n)$ as follows:

$$\varphi(x_1 x_2 \cdots x_m) = \varphi(x_1) \varphi(x_2) \cdots \varphi(x_m),$$

$$x_i \in \{a, b\}$$

$$\varphi(a) = b, \quad \varphi(b) = ab, \quad f_0 = a,$$

$$f_n = \varphi((f_{n-1})^R).$$

The following words will be generated by the function $\varphi(f_n)$:

$$f_0 = a$$

$$f_1 = \varphi((f_0)^R) = \varphi(a) = b$$

$$f_2 = \varphi((f_1)^R) = \varphi(b) = ab$$

$$f_3 = \varphi((f_2)^R) = \varphi((ab)^R) = \varphi(ba) = abb$$

$$f_4 = \varphi((f_3)^R) = \varphi((abb)^R) = \varphi(bba) = ababb$$

$$f_5 = \varphi((f_4)^R) = \varphi((ababb)^R) = \varphi(bbaba) = ababbabb$$

$$f_6 = \varphi((f_5)^R) = \varphi((ababbabb)^R) = \varphi(bbabbaba) = ababbababb$$

Property 3.1. $\varphi(f_n)$ generates alternating Fi-

bonacci words.

Proof. Essentially, we want to prove that

$$\begin{cases} f_n = f_{n-2} + f_{n-1} & \text{if } n \text{ is even} \\ f_n = f_{n-1} + f_{n-2} & \text{if } n \text{ is odd.} \end{cases} \quad (1)$$

Note that Equation (1) is true for $n = 2, 3, 4$, because $f_2 = ab = f_0 + f_1$, $f_3 = abb = f_2 + f_1$, $f_4 = ababb = f_2 + f_3$.

Assume that

$$\begin{cases} f_i = f_{i-2} + f_{i-1} & \text{if } i \text{ is even} \\ f_i = f_{i-1} + f_{i-2} & \text{if } i \text{ is odd,} \end{cases} \quad \text{for } i = 2, 3, \dots, n.$$

We now show that

$$\begin{cases} f_{i+1} = f_{i-1} + f_i & \text{if } i+1 \text{ is even} \\ f_{i+1} = f_i + f_{i-1} & \text{if } i+1 \text{ is odd.} \end{cases}$$

Consider $i+1$ is odd, i will be even.

$$\begin{aligned} f_{i+1} &= \varphi((f_i)^R) \\ &= \varphi((f_{i-2} + f_{i-1})^R) \\ &= \varphi((f_{i-1})^R) + \varphi((f_{i-2})^R) \\ &= f_i + f_{i-1} \end{aligned}$$

Similarly, $f_{i+1} = f_{i-1} + f_i$, if $i+1$ is even. $\square$

We give several other functions in Table 2 which also generate alternating Fibonacci words.

Table 2: The Other Alternating Fibonacci Word Generation Functions

φ_1	$f_0 = a$	$\varphi_1(a) = b$ $\varphi_1(b) = ba$	$f_n = \varphi_1((f_{n-1})^R)$
φ_2	$f_0 = a$	$\varphi_2(a) = ab$ $\varphi_2(b) = a$	$f_n = \varphi_2((f_{n-1})^R)$
φ_3	$f_0 = a$	$\varphi_3(a) = ba$ $\varphi_3(b) = a$	$f_n = \varphi_3((f_{n-1})^R)$

4 Block Palindrome

In this section, we shall introduce a new term, the block reverse concept: Let x and y be two strings, where $x = x_1 + x_2$, $y = y_1 + y_2$ and $|x_1| = |x_2| = |y_1| = |y_2|$. y is a block reverse of x if $y_1 = x_2$ and $y_2 = x_1$. For instance, $y = ba$ is a block reverse of $x = ab$. Besides, $y = baab$ is a block reverse of $x = abba$. We shall use x^{BR} to denote the block reverse of x .

Having defined block reverse, we shall define the block palindrome as follows: If $|S| = 4$, then S is a block palindrome if and only if S is a palindrome. If $|S| = 2^n > 4$, S is a block palindrome if and only if $S = S_1 + S_2$, $|S_1| = |S_2|$, S_2 is a block reverse of S_1 and both S_1 and S_2 are block palindromes.

For instance, According to our definition, $S = abbabaabbaababba$ is a block palindrome, because S can be divided into $S_1 = abbabaab$ and $S_2 = baababba$, $|S_1| = |S_2|$, S_2 is a block reverse of S_1 , and S_1 and S_2 are both block palindromes.

We propose a word generating function $\chi(f_n)$ as follows:

$$\chi(x_1 x_2 \cdots x_m) = \chi(x_1) \chi(x_2) \cdots \chi(x_m), \quad x_i \in \{a, b\}$$

$$\begin{aligned} \chi(a) &= ab, \quad \chi(b) = ba, \quad f_0 = a, \\ f_n &= \chi(f_{n-1}). \end{aligned}$$

The following words will be generated by the function $\chi(f_n)$ and all are block palindrome:

$$\begin{aligned} f_0 &= a \\ f_1 &= \chi(f_0) = \chi(a) = ab \\ f_2 &= \chi(f_1) = \chi(ab) = abba \\ f_3 &= \chi(f_2) = \chi(abba) = abbabaab \\ f_4 &= \chi(f_3) = \chi(abbabaab) = abbabaabbaababba \\ f_5 &= \chi(f_4) = \chi(abbabaabbaababba) \\ &= abbabaabbaababbabaababbaababbaab \end{aligned}$$

Property 4.1. $\chi(f_n)$ generates block palindrome words.

Proof. We want to prove that

$$f_n = f_{n-1} + (f_{n-1})^{BR} \quad \text{for } n > 1 \quad (2)$$

Note that Equation (2) is valid for $n = 2, 3, 4$,

because

$$\begin{aligned} f_2 &= abba = ab + ba = f_1 + (f_1)^{BR} \\ f_3 &= abbabaab = abba + baab = f_2 + (f_2)^{BR} \\ f_4 &= abbabaabbaabba \\ &= abbabaab + baababba = f_3 + (f_3)^{BR} \end{aligned}$$

Assume that

$$f_i = f_{i-1} + (f_{i-1})^{BR} \quad \text{for } i = 2, 3, \dots, n \text{ are true.}$$

We want to show

$$f_{i+1} = f_i + (f_i)^{BR}$$

Consider,

$$\begin{aligned} f_{i+1} &= \chi(f_i) \\ &= \chi(f_{i-1} + (f_{i-1})^{BR}) \\ &= \chi(f_{i-1}) + \chi((f_{i-1})^{BR}) \\ &= f_i + \chi((f_{i-1})^{BR}) \end{aligned} \quad (3)$$

Next, we will show that $\chi((f_{i-1})^{BR}) = (f_i)^{BR}$.

Let us consider

$$f_i = f_{i-1} + (f_{i-1})^{BR} \quad (4)$$

$$\begin{aligned} f_i &= \chi(f_{i-1}) \\ &= \chi(f_{i-2} + (f_{i-2})^{BR}) \\ &= \chi((f_{i-2}) + \chi((f_{i-2})^{BR})) \\ &= f_{i-1} + \chi((f_{i-2})^{BR}) \end{aligned} \quad (5)$$

By (4) and (5), we have $(f_{i-1})^{BR} = \chi((f_{i-2})^{BR})$.

Then, by definition of block reverse, we have

$$(f_{i-1})^{BR} = (f_{i-2})^{BR} + f_{i-2}.$$

$$\begin{aligned} \chi((f_{i-1})^{BR}) &= \chi((f_{i-2})^{BR} + f_{i-2}) \\ &= \chi((f_{i-2})^{BR}) + \chi(f_{i-2}) \\ &= \chi((f_{i-2})^{BR}) + \chi(f_{i-2}) \\ &= (f_{i-1})^{BR} + f_{i-1} \\ &= (f_i)^{BR} \end{aligned} \quad (6)$$

Hence, by(3) and (6), we have $f_{i+1} = f_i + (f_i)^{BR}$. $\square$

5 The Suffix-Prefix Fibonacci Words

In this section, we introduce a new word generation function, suffix-prefix Fibonacci words. The word generating function $\xi(w_n)$ is as follows:

$$\begin{aligned} \xi(x_1 x_2 \dots x_m) &= \xi(x_1) \xi(x_2) \dots \xi(x_m), \quad x_i \in \{a, b\} \\ \xi(a) &= ab, \quad \xi(b) = bc, \quad \xi(c) = ca, \quad w_0 = a, \\ w_n &= \xi(w_{n-1}). \end{aligned}$$

The following words can be generated by the above definition:

$$\begin{aligned} w_0 &= a \\ w_1 &= \xi(w_0) = \xi(a) = ab \\ w_2 &= \xi(w_1) = \xi(ab) = abbc \\ w_3 &= \xi(w_2) = \xi(abbc) = abbcbbcca \\ w_4 &= \xi(w_3) = \xi(abbcbbcca) = abbcbbccabccacaab \\ w_5 &= \xi(w_4) = \xi(abbcbbccabccacaab) \\ &= abbcbbccabccacaabccacaabcaababbc \end{aligned}$$

Before we describe the suffix-prefix Fibonacci

words, we define some notations as follows:

$|w|$: The length of the word w .

$w^{\frac{pre}{k}}$: The prefix of the word whose length is $\frac{|w|}{k}$.

$w^{\frac{suf}{k}}$:: The suffix the word w whose length is $\frac{|w|}{k}$.

A suffix-prefix Fibonacci words is a word which satisfies the following condition:

$$w_n = w_{n-1} + w_{n-1}^{\frac{suf}{2}} + w_{n-1}^{\frac{suf}{4}} + w_{n-1}^{\frac{pre}{4}} \quad \text{for } n \geq 3$$

. For example:

$$\begin{aligned} w_3 &= abbcacca \\ w_4 &= w_3 + w_3^{\frac{su f}{2}} + w_3^{\frac{su f}{4}} + w_3^{\frac{pre}{4}} \\ &= abbcacca + bcca + ca + ab \\ &= abbcaccabccacaab \end{aligned}$$

The generating function $\xi(f_n)$ has the following properties:

Property 5.1. $\xi(f_n)$ generates suffix-prefix Fibonacci words.

Proof. First, we consider $n=3$.

$$\text{Because } w_3 = abbcacca = abbc + bc + c + a = w_2 + w_2^{\frac{su f}{2}} + w_2^{\frac{su f}{4}} + w_2^{\frac{pre}{4}}$$

Therefore, $n = 3$ holds.

Assume $n=k$ holds. That is,

$$w_k = w_{k-1} + w_{k-1}^{\frac{su f}{2}} + w_{k-1}^{\frac{su f}{4}} + w_{k-1}^{\frac{pre}{4}}$$

Let us consider $n=k+1$,

$$\begin{aligned} w_{k+1} &= \xi(w_k) \\ &= \xi(w_{k-1} + w_{k-1}^{\frac{su f}{2}} + w_{k-1}^{\frac{su f}{4}} + w_{k-1}^{\frac{pre}{4}}) \\ &= \xi(w_{k-1}) + \xi(w_{k-1}^{\frac{su f}{2}}) + \xi(w_{k-1}^{\frac{su f}{4}}) + \xi(w_{k-1}^{\frac{pre}{4}}) \\ &= w_k + w_k^{\frac{su f}{2}} + w_k^{\frac{su f}{4}} + w_k^{\frac{pre}{4}} \end{aligned}$$

Thus, we proved. $\square$

Property 5.2. $|w_n| = 2|w_{n-1}|$

$$\text{Example: } w_3 = abbcacca, |w_3| = 8, w_4 = abbcaccabccacaab, |w_4| = 16.$$

Property 5.3. $|w_n| = 2^n$

$$\text{Example: } w_4 = abbcaccabccacaab, |w_4| = 2^4 = 16.$$

Property 5.4. $w_n^{\frac{pre}{2k}} = \xi(w_{n-1}^{\frac{pre}{2k}})$ where $k = 1, 2, \dots, n-1$

$$\begin{aligned} \text{Example: } w_3 &= abbcacca, w_4 = abbcaccabccacaab, \\ w_3^{\frac{pre}{4}} &= ab, w_4^{\frac{pre}{4}} = \xi(ab) = abbc \end{aligned}$$

Property 5.5. $w_n^{\frac{su f}{2k}} = \xi(w_{n-1}^{\frac{su f}{2k}})$ where $k = 1, 2, \dots, n-1$

$$\begin{aligned} \text{Example: } w_3 &= abbcacca, w_4 = abbcaccabccacaab, \\ w_3^{\frac{su f}{4}} &= ca, w_4^{\frac{su f}{4}} = \xi(ca) = caab \end{aligned}$$

6 Conclusions

In this paper, we propose some new word generation functions and study his properties. These functions include Fibonacci words, alternating Fibonacci words, the block palindrome words and the suffix-prefix Fibonacci words.

Some new applications of generating functions on algorithmic art are very interesting. In the future, we will apply these generating function to the algorithmic art, such as music or visual art.

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